

0 8 . 1

State **and** explain the property of water that can help to buffer changes in temperature.

[2 marks]

0 8 . 2

Water is used to hydrolyse ATP.

Name the **two** products of ATP hydrolysis.

[1 mark]

1

2

Turn over ►

Hydrolysis of ATP is catalysed by the enzyme ATP hydrolase.

A student investigated the effect of ATP concentration on the activity of ATP hydrolase. She used shortening of strips of muscle tissue caused by contraction as evidence that ATP was being hydrolysed.

- She took four slides **A**, **B**, **C** and **D**, and added strips of muscle tissue of the same length to each slide.
- She then added the same volume of ATP solutions of different concentrations to the four slides and left each slide for five minutes.
- She then recorded the final length of each strip of muscle tissue.

Her results can be seen in **Table 1**.

Table 1

Slide	Concentration of ATP solution added to slide / $\times 10^{-6} \text{ mol dm}^{-3}$	Final length of muscle tissue after 5 minutes / mm
A	2	36
B	4	31
C	6	29
D	8	26

0 8 . 3

Other than those given, name **two** variables the student should have controlled.

[2 marks]

1 _____

2 _____



0 8 . 4

Describe **and** explain the pattern shown by the data in **Table 1**.**[2 marks]**

Description _____

Explanation _____

0 8 . 5

The hydrolysis of 1 dm^3 of a 1 mol dm^{-3} solution of ATP releases 30 500 J of energy.60% of the energy released during the hydrolysis of 1 mol dm^{-3} of ATP is released as heat; the rest is used for muscle contraction.The student added 0.05 cm^3 of ATP solution to slide **D**.

Calculate the energy available from ATP for contraction of the muscle on this slide.

[3 marks]

Answer = _____ J

10

Turn over ►

